

MLOps Pipeline for Bird Species Identification from Audio

Group 2:

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Machine Learning and Data in Operation

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Components and Tools



**Data Versioning & Model
Registry**



**Automated Pipelines /
Workflows**



**Model Training, Evaluation &
Inference**



Deployment



 **Label Studio**

Labeling

Data Versioning & Tracking



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- **Reproducibility:** DVC tracks versions of files and datasets alongside Git-managed code, allowing reproducibility of experiments.
 - **Storage Efficiency:** Large audio files are efficiently stored externally keeping the Git repository lightweight.
 - **Collaboration & Transparency:** Development integration with Git for collaboration and visibility across the team.

Automated Pipeline



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- **Centralized Experiment Tracking:** A local MLFlow server instance to log all metrics, making our project easily reproducible and transparent
 - **Effective Model Registry:** Manage multiple model versions and deployment stages providing governance and ensuring smooth transitions between development, staging, and production
 - **Integrated Workflow:** MLFlow provides easy integration with all key components and tools we selected in our pipeline

Training and Evaluation

 PyTorch  ONNX

- **Training:** pre-trained backbone to account for data scarcity, log-mel and SpecAugment the Kaggle bird audio files; mixed-precision **finetuning** on a single GPU for efficiency
- **Evaluation:** track with MLFlow train/val/test class and average loss and perplexity curves, as well as ROC AUC required by the competition
- **Inference:** Serve with onnxruntime for low-latency inference

API Deployment & Monitoring



- Dockerized environment for stability, reproducibility, and deployment **flexibility** on heterogeneous client setups
- FastAPI provides lightweight, high-speed endpoints ideal for model serving in production, supporting **fast predictions** and easy CI/CD integration
- **Built-in Monitoring:** Confidence scores route predictions reliable to users uncertain to labeling

Labeling



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- **Human-in-the-loop** feedback embedded in the design of the pipeline, to improve data quality and model accuracy
 - **API integration** between Label Studio and our API, to incentivize and simplify data collection efforts
 - **Automated** versioning and storing of the newly labeled audio files, to be ready for the next improvement cycle